

Package ‘geometry’ detected.This can cause a problem for jmlrbook

Project 1 - Explainable AI in Radiology

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Abstract

This study applies explainable AI techniques to detect pleural effusion in chest X-rays using a DenseNet169 model. Saliency Maps highlight image regions relevant to predictions, while Example-Based Explanations retrieve similar cases for comparison. Explanation quality is assessed using Quantus for Saliency Maps and Normalized Discounted Cumulative Gain (nDCG) for case rankings. The explanations presented in this work show good initial results but would still require improvement.

Keywords: Chest X-Ray, Pleural Effusion, Explainable AI, Saliency maps, GradCAM, Example-based explanations, Cosine Similarity, nDCG, DenseNet

1. Introduction

Machine learning is increasingly used to solve complex problems across domains. In healthcare, particularly in radiology imaging, AI supports diagnostic and treatment decisions (7). However, many models act as black boxes with limited insight into how they generate outputs (2), making explainable AI essential to improve transparency and trust.

X-ray analysis is one of the main applications of AI in medicine, especially for detecting bone and lung diseases (9).

This paper presents a machine learning model to predict pleural effusion from CXR images. It evaluates two explanation methods: Saliency maps and Cosine Similarity. Their quality is evaluated using Quantus (8) and the Normalized Discounted Cumulative Gain (nDCG) score.

2. Methods & Materials

Dataset and Preprocessing

The dataset used for this study was a smaller version of the CheXpert dataset (?), containing frontal and lateral chest X-rays, of which only the frontal images were used. Among the pathology labels available, particular attention was paid to *Pleural Effusion*, which served as the classification target.

Images with missing or uncertain *Pleural Effusion* labels were removed to improve reliability and all images were resized to 224×224 pixels and normalized to the $[0, 1]$ range. To address the class imbalance, class weights were calculated based on label frequencies and applied during training, where an 80-20 train-test-split was used. Since the original X-rays were grayscale, each image was duplicated across three channels to match the input format required by ImageNet-pretrained models.

Model Architecture

For the classification of *Pleural Effusion*, a convolutional neural network (CNN), the **DenseNet169** (?) was applied.

It consists of 169 layers organized into densely connected blocks separated by transition layers that simplify spatial dimensions for efficiency. It strikes a balance between model depth and computational

cost, making it suitable for medical imaging, where subtle patterns require deep, feature-rich networks (?).

The architecture was adapted from an existing implementation (?) and initialized with ImageNet (?) pre-trained weights, which capture general low-level features useful for chest X-rays. During training, the lower layers were frozen and the upper layers fine-tuned to detect *Pleural Effusion*.

Saliency Map-based Explanations

One form of visual explanation for deep learning models is saliency map-based explanations. This report specifically explores the use of Gradient-weighted Class Activation Mapping (**Grad-CAM**) which uses gradients in the last convolutional layer of the CNN, and global-average-pooling to determine each neuron’s importance to the final prediction (10), and can be used to generate heatmaps to highlight critical information in the image.

Additionally, to evaluate the quality of saliency maps, **Quantus** (8), a toolkit for evaluating deep learning explanations, was used. Specifically, it applies the following metrics: Pixel-Flipping (5) for Faithfulness, Local Lipschitz Estimate (3) for Robustness, Model Parameter Randomisation test (1) for Randomisation, Relevance Mass Accuracy (4) for Localisation, and Complexity (6) for Complexity.

Example-based Explanations

Example-based explanations are performed by extracting image feature embeddings from the last pooling layer and ranking images by **cosine similarity**, which measures vector orientation and magnitude (?). Those rankings are compared to expert-defined rankings using **nDCG**, which evaluates ranking quality by assigning relevance scores (here in the range [5.5, 1]), to compute discounted gain, and normalizing it based on the ideal expert ranking (?). This approach evaluates whether the model groups similar images in a clinically meaningful way.

3. Results & Discussion

In the subsequent sections, we will present and discuss the results gathered from testing the model and explanation methods described in section 2 on a predefined set of test images.

Model Training and Evaluation

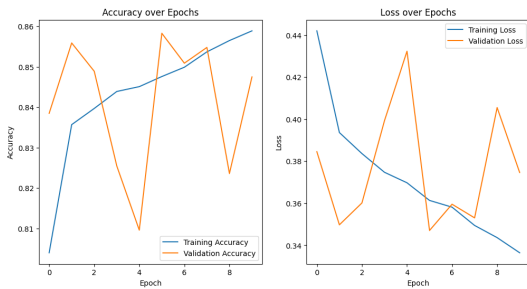


Figure 1: Training and validation performance per epoch

During training, the DenseNet169 model steadily improved in training accuracy and loss, though validation accuracy fluctuated, indicating some overfitting (Figure 1). The final model achieved 79.21% accuracy on the test set, with a precision of 63.41% and recall of 81.25%, showing good detection of positive *Pleural Effusion* cases. The F1-score was 71.23%, balancing precision and recall, while the ROC AUC reached 84.96%, demonstrating strong overall discriminatory power. A binomial test yielded a p-value of 2.03×10^{-17} , confirming the performance is significantly better than chance.

Saliency Maps

Figure 3 shows the GradCAM explanations generated to gain insights about the models’ focus on the test images. The most important region, as identified by the model, is represented in seismic colours: red, white, and blue. The bounding boxes on the explanation boxes circle the critical parts of the image, that the explanation should focus on.

In all five test images, the explanations highlight parts of the bounding box but often highlight irrelevant areas outside of it.

In most cases, the heatmaps focus on the lower lung regions associated with Pleural Effusion, suggesting the model identifies clinically relevant areas. However, some explanations highlight unrelated regions, indicating that the model occasionally relies on uninformative features to classify Pleural Effusion.

Additionally, table 3 shows the mean and standard deviation of the Quantus evaluation metrics on the test images ¹.

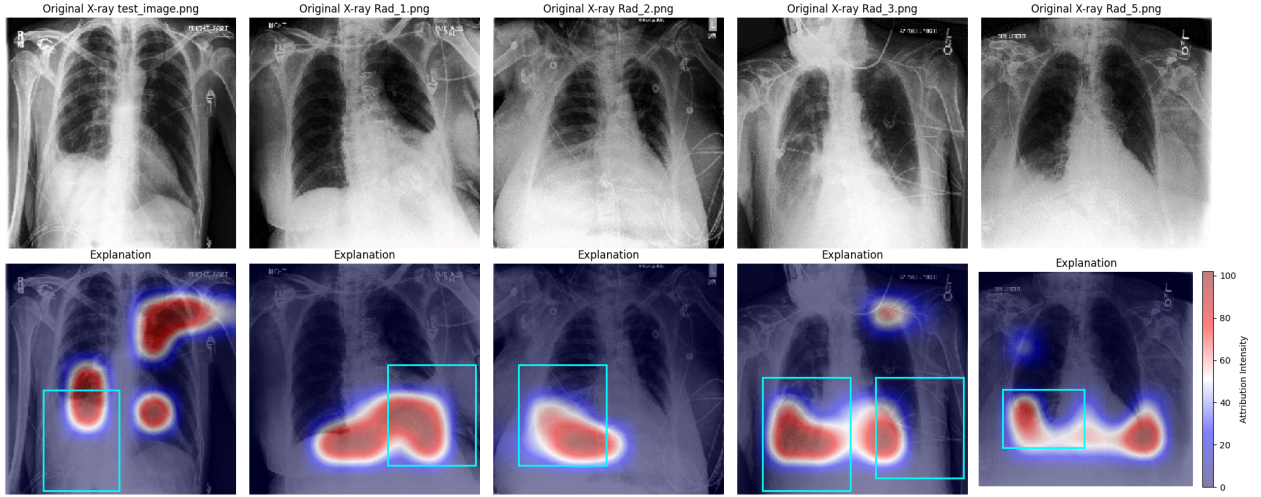


Figure 2: GradCAM Explanations

	Mean	Standard deviation
Complexity	9.360562	0.241967
Local Lipschitz Estimate	0.032051	0.031284
Pixel Flipping	1	0

Figure 3: Quantus: Explanation Evaluation

The complexity metric (9.3) reflects the expected high complexity of CNN explanations. The robustness score (0.032051) indicates the model is highly stable to small input perturbations. Finally, the faithfulness measure confirms that removing important pixels causes a significant drop in confidence, showing the explanation accurately highlights the model’s key features.

Example-based Explanations

The results indicate an alignment between the scientists’ and the cosine similarity rankings. The high nDCG score of 0.96 means that features of the images extracted by the model exhibit relevant information for accurate classification that correlates the opinion of the scientist. While the line plot 4 reveals some mismatches in rankings, the overall similarity remains consistent; the cosine similarity score of the images

1. Please note that the other two metrics (Randomisation and Localisation) were left out of the table due to practical errors between the TensorFlow model and the Quantus explanations, which is explained in Appendix A.

in the same ranking spot in different lines tends to have close values. Especially in the first half of the rankings compared to the second half which, given the discounted nature of the evaluation metric, affect the high nDCG score. The heatmap in plot 5, shows that both rankings have the same five images in their top six rankings in slightly different order.

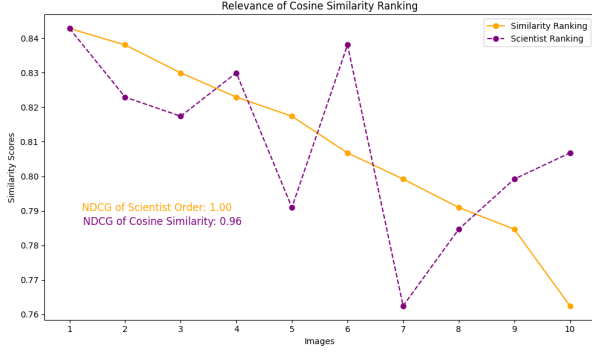


Figure 4: The line plot of images ranked by the scientists and cosine similarity.

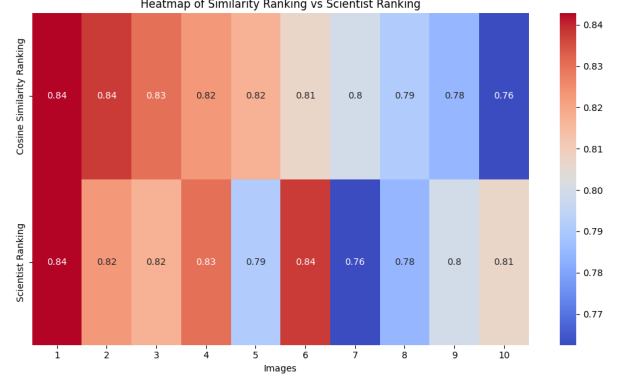


Figure 5: The heatmap of the rankings (identical images share the same color)

4. Conclusion

The understanding of a deep neural network decision-making process is important to make an explainable system. In this study, multiple methods are used to check the explainability of the model we created. Overall results showed promising outcomes, especially having a close ranking compared to a professional. However, there are some improvements to be made in the future to improve the performance and consistency of the model.

Appendix A. Quantus Error Messages

While Quantus was adapted to TensorFlow, it appears that two of the requested methods are not fully compatible. We have looked for help but could not fix it until the deadline and had not trained a new model since we were initially convinced, that with the promise of compatibility, the method should work, at least with the in-built explanation methods of quantus - which it did not. In Figure 6 one can see the error displayed when applying the localisation metric (Relevance Mass Accuracy) to the in-built explanation method of Integrated Gradients. The inputs should share same dimensionserror presents itself even when the shapes of the two specified inputs are clearly the same.

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Figure 6: The error displayed when for the localisation metric on the in-built Integrated Gradients explanation method with a TensorFlow model.

```

[50] print(x_batch.shape == s_batch.shape)
True

[51] # 5. Localisation: relevance mass accuracy (Arras et al., 2021)
localisation = quantus.RelevanceMassAccuracy(
)(model=model,
  x_batch=x_batch,
  y_batch=y_predicted,
  a_batch=None,

  #explain_func= custom_explain,
  explain_func=quantus.explain,
  explain_func_kwargs={"method": "IntegratedGradients"},
  s_batch=s_batch,
)

print(f"Localisation Scores (relevance mass accuracy): {localisation}")

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AssertionError                                Traceback (most recent call last)
<ipython-input-51-74a0a3fbf4b3> in <cell line: 0>()
      1 # 5. Localisation: relevance mass accuracy (Arras et al., 2021)
----> 2 localisation = quantus.RelevanceMassAccuracy(
      3 )(model=model,
      4     x_batch=x_batch,
      5     y_batch=y_predicted,

----- 4 frames -----
/usr/local/lib/python3.11/dist-packages/quantus/helpers/asserts.py in assert_segmentations(x_batch, s_batch)
    219 ), "The inputs 'x_batch' and segmentations 's_batch' should include the same number of samples."
    220 assert (
--> 221     np.shape(x_batch)[2:] == np.shape(s_batch)[2:]
    222 ), "The inputs 'x_batch' and segmentations 's_batch' should share the same dimensions."
    223 assert (

AssertionError: The inputs 'x_batch' and segmentations 's_batch' should share the same dimensions.

```

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